



Does [dak] mean [ɫɫk]?:

Infants recognize many other-accented words prior to 19 months

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Background

- Before the age of 19 months, infants have failed to spontaneously adapt to new accents in lab-based tasks (Best et al., 2009; Mulak et al., 2013; van Heugten & Johnson, 2014) – leading to a **widely-accepted view that young infants *can't* adapt**.
- Tested items in these studies feature **systematic segmental changes** between accents (i.e. vowel shifts that cross perceptual category boundaries, rhoticity).
- Across natural accents, many words may **lack such segmental differences**. Spontaneous recognition of these words may be substantially easier for infants.

Are young infants able to recognize *some* words in unfamiliar accents?



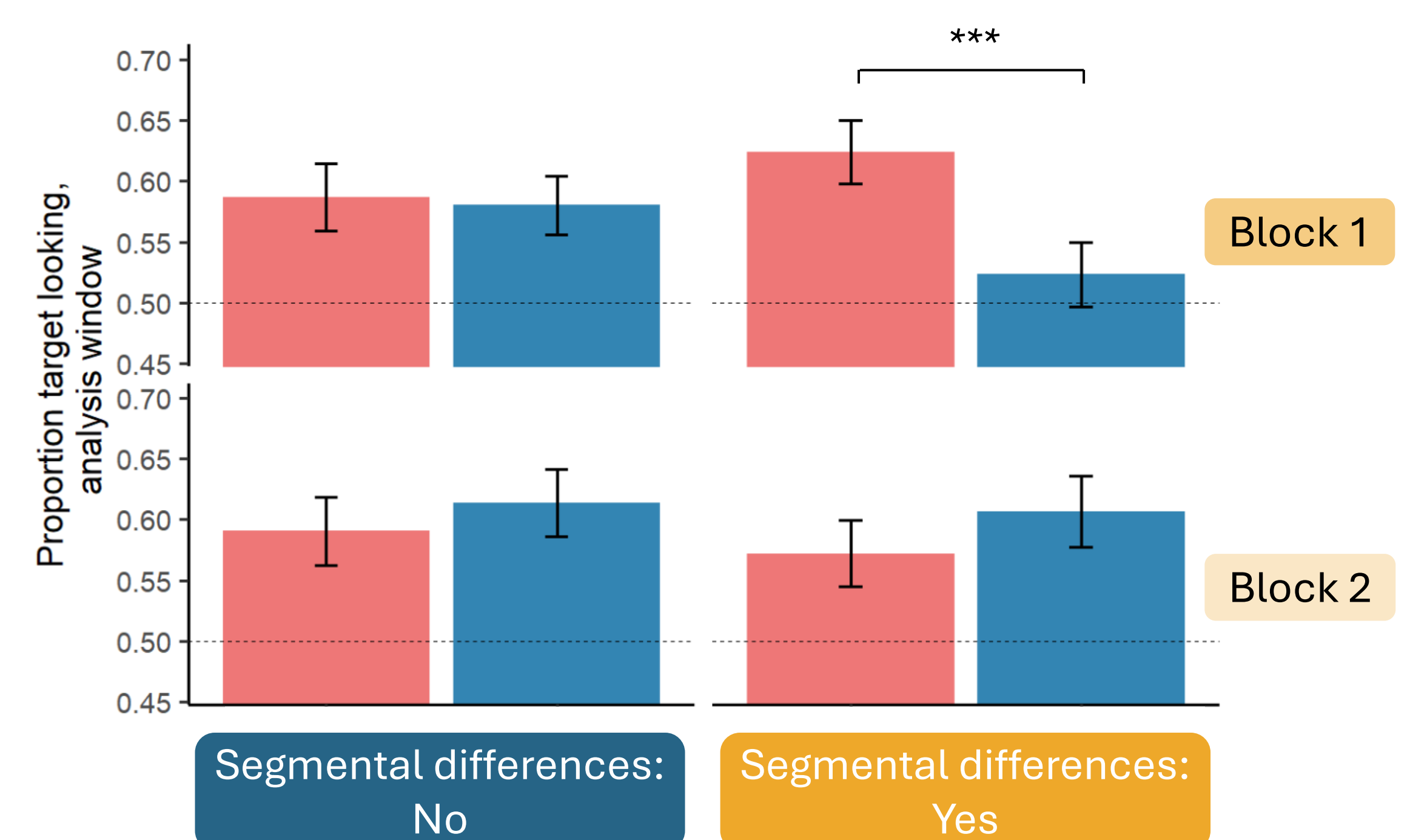
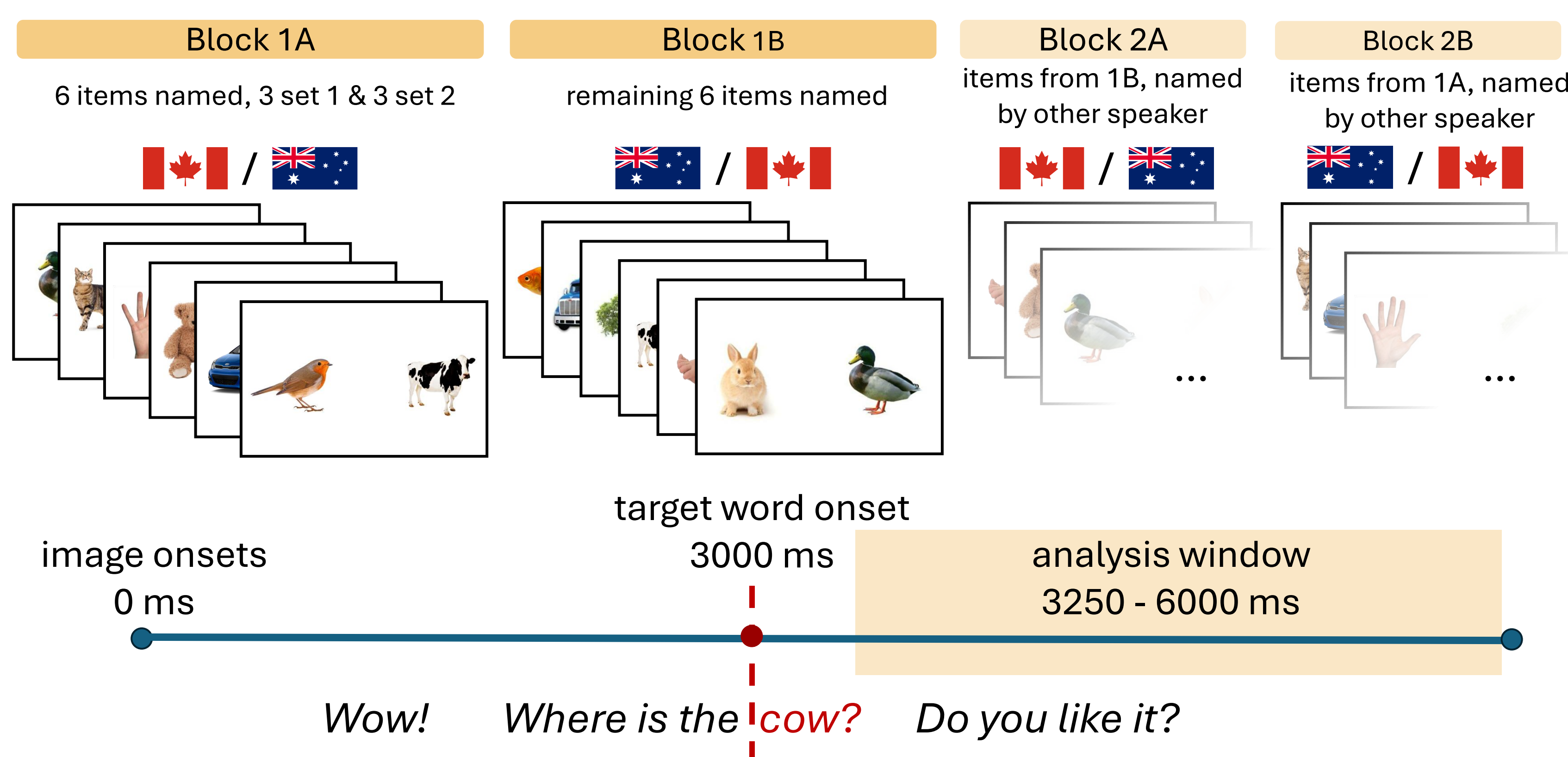
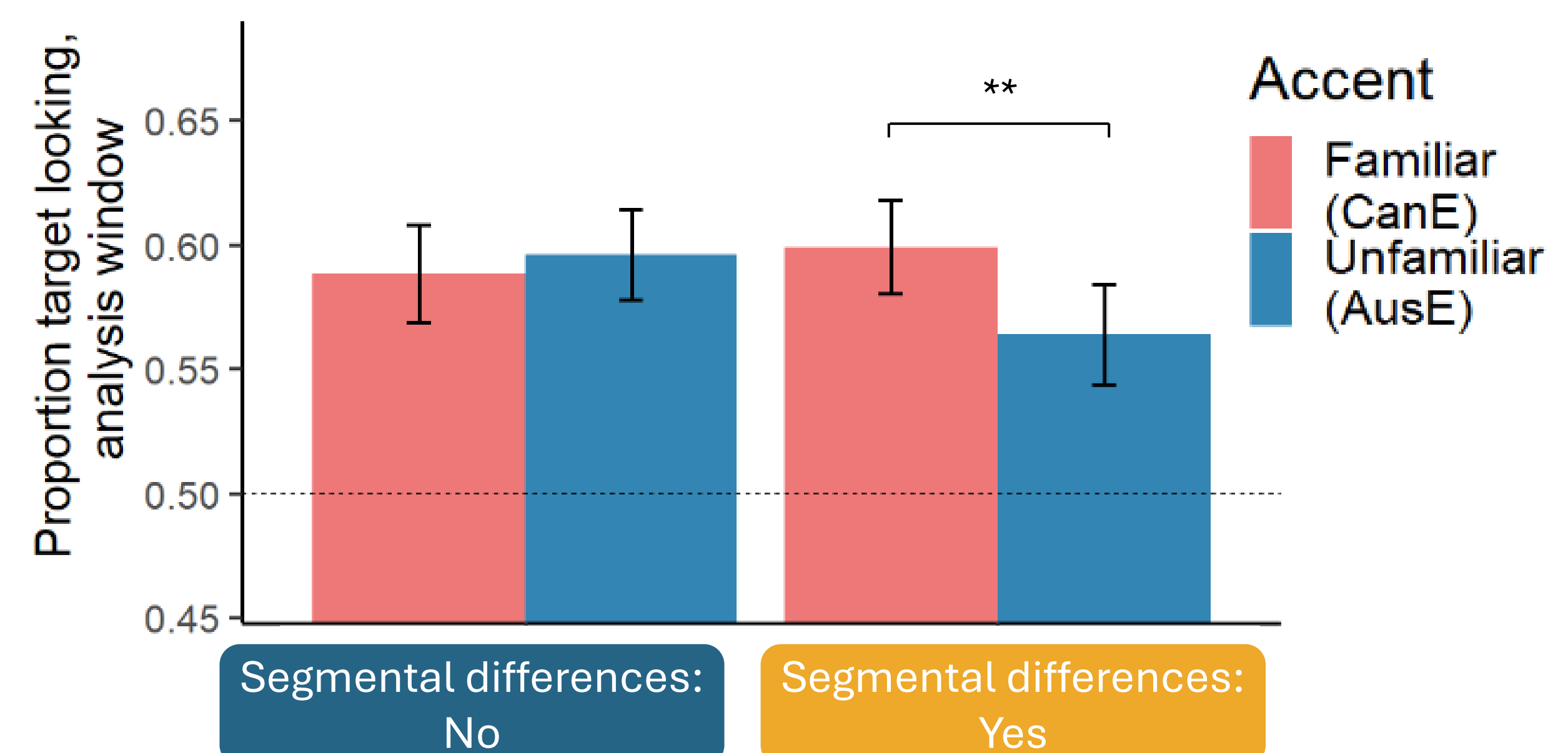
Study 1: Are segmental changes across accents particularly difficult for infants?

Participants: 16-month-olds ($n = 40$)

Stimuli: Items selected for pronunciation across **familiar (Canadian English/CanE)** and **unfamiliar (Australian English/AusE)** accents

[†]Wordbank: Frank et al., 2016

Item set 1: No segmental differences				Item set 2: Have segmental differences			
Word	CanE	AusE	Vocab norm [†]	Word	CanE	AusE	Vocab norm [†]
bunny	bʌni	bʌni	0.66	bear	bɛəɹ	bɛ:ə	0.64
cat	kʰæt̚	kʰæt̚	0.84	bird	bɝd	bɜ:d	0.82
cow	kʰaw	kʰaw	0.69	car	kʰɑɹ	kʰɑ:	0.91
foot	fʊt	fʊt	0.83	duck	dʌk	dak	0.79
hand	hænd	hænd	0.73	fish	fɪʃ	fɪjʃ	0.66
tree	tʃi:	tʃi:	0.59	truck	tʃɹʌk	tʃɹɛk	0.65
average			0.72	average			0.75



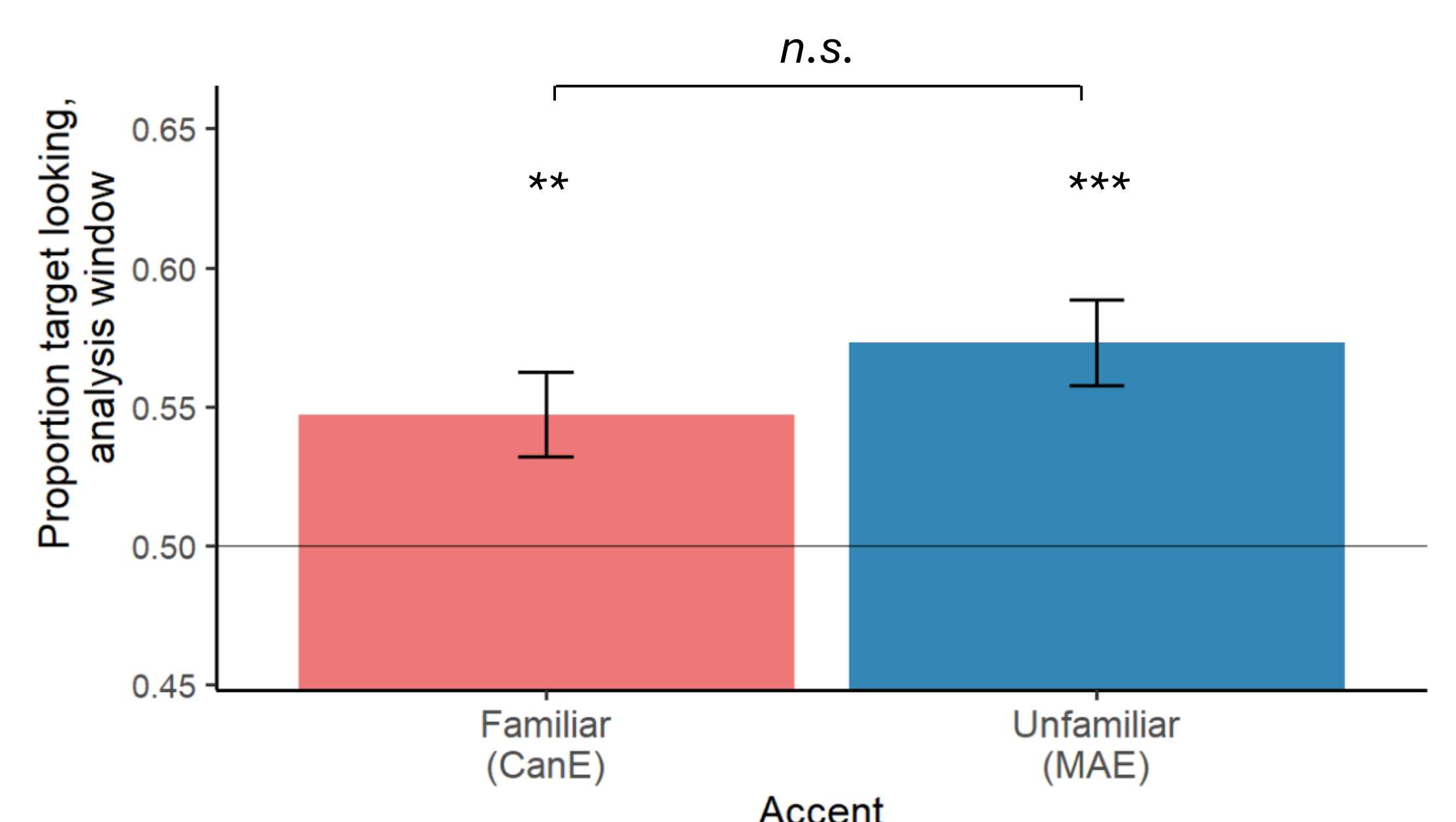
Study 2: Is cross-accent word recognition possible even earlier in infancy?

Participants: 14-month-olds ($n = 37$)

Stimuli: Items chosen to minimize segmental changes between **familiar (CanE)** and **unfamiliar (Mandarin-accented English/MAE)** accents

Task: Eyetracking task paralleling Study 1, without item set distinction

Items							
Word	CanE	MAE	Vocab norm [†]	Word	CanE	MAE	Vocab norm [†]
baby	beɪbi	beɪbi	0.77	dog	dʌg	dak	0.83
ball	bɑ:l	bɑ:l	0.90	fish	fɪʃ	fɪjʃ	0.53
bird	bɝd	bɝd	0.65	foot	fʊt	fʊt	0.62
car	kʰɑɹ	kʰɑɹ	0.78	hand	hænd	hænd	0.54
cat	kʰæt̚	kʰæt̚	0.67	hat	hæt̚	hæt̚	0.55
cow	kʰaw	kʰaw	0.51	shoe	ʃu:w	ʃu:w	0.82
average			0.58				



Discussion

- Contrary to previous reports, infants as young as 14 months spontaneously understand many other-accented words.
- Further work will target when this ability emerges, and whether recognition is limited to words that are segmentally-matched across accents, or if it extends to segmentally-different words (see Study 1, block 2).
- If current trends hold, this could support claims that abstract phonological knowledge is early-emerging (Johnson & White, 2025).

